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B.Tech/M.Tech (Integrated) DEGREE EXAMINATION, MAY 2025

Sixth Semester

21ECC302T - ANALOG AND DIGITAL COMMUNICATION

(For the candidates admitted during the academic year 2021-2022 to 2024-2025)

Note:

- i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 × 1 = 20 Marks)

Answer **all** Questions

Marks BL CO

- | | | | |
|---|---|---|---|
| <p>1. A product modulator produces a _____ signal.
 (A) AM-FC (B) DSB-SC
 (C) SSB (D) VSB</p> | 1 | 2 | 1 |
| <p>2. In a critically modulated AM system, if the total power is 150W, what is the power in carrier?
 (A) 225W (B) 150W
 (C) 100W (D) 50W</p> | 1 | 3 | 1 |
| <p>3. A 15KHz single tone signal is frequency modulated with modulation index of 5. Find the bandwidth using Carson's rule.
 (A) 75 KHz (B) 90 KHz
 (C) 150 KHz (D) 180 KHz</p> | 1 | 2 | 1 |
| <p>4. Figure of merit is always unity in _____.
 (A) FM (B) PM
 (C) SSB-SC (D) AM-FC</p> | 1 | 1 | 1 |
| <p>5. The main advantage of TRF receiver is _____.
 (A) high Selectivity (B) high Sensitivity
 (C) simple hardware (D) low interference</p> | 1 | 2 | 2 |
| <p>6. The image frequency of an AM radio station with carrier frequency 580 KHz is _____.
 (A) 725 KHz (B) 935 KHz
 (C) 1160 KHz (D) 1490 KHz</p> | 1 | 2 | 2 |
| <p>7. Why is FM radio broadcasting has good quality than AM radio broadcasting?
 (A) has more sidebands (B) uses more bandwidth
 (C) has more noise immunity (D) uses LoS propagation</p> | 1 | 2 | 2 |
| <p>8. The intermediate frequency used in commercial FM Radio broadcasting receiver is _____ MHz.
 (A) 4.5 (B) 10.7
 (C) 107.8 (D) 87.7</p> | 1 | 1 | 2 |
| <p>9. The maximum permissible sampling interval of a 4 KHz signal is _____ microsecond.
 (A) 8000 (B) 4000
 (C) 1000 (D) 125</p> | 1 | 2 | 3 |
| <p>10. In PCM, if the quantization level is increased from 16 to 256, the bandwidth requirement will increase by _____ times.
 (A) 2 (B) 4
 (C) 8 (D) 16</p> | 1 | 3 | 3 |

- | | | | |
|--|-------------------------------------|---|---|
| 11. The signal to quantization noise ratio of delta modulator for a 4 KHz signal sampled at 64 KHz without slope overload is approximately given by _____ dB. | 1 | 4 | 3 |
| (A) 16 | (B) 22 | | |
| (C) 32 | (D) 44 | | |
| 12. A speech signal with 10V peak-to-peak voltage range is quantized with 265 levels. The rms value of quantization noise is approximately _____ millivolts. | 1 | 3 | 3 |
| (A) 11.3 | (B) 12.7 | | |
| (C) 15.3 | (D) 39.0 | | |
| 13. Which of the following yields maximum probability of error? | 1 | 2 | 4 |
| (A) ASK | (B) FSK | | |
| (C) PSK | (D) DPSK | | |
| 14. For a bit rate of 8 Kbps, the best possible values of BFSK carrier frequencies are _____ and _____. | 1 | 2 | 4 |
| (A) 16 KHz, 20 KHz | (B) 20 KHz, 24 KHz | | |
| (C) 20 KHz, 32 KHz | (D) 32 KHz, 40 KHz | | |
| 15. If binary PSK modulation needs a minimum bandwidth of 9.6 KHz, what modulation scheme to be adopted to reduce the bandwidth to 2.4 KHz? | 1 | 2 | 4 |
| (A) DPSK | (B) QPSK | | |
| (C) 8-ary PSK | (D) 16-ary QAM | | |
| 16. In a BPSK signalling, the received carrier amplitude is 1V, and the PSD of AWGN at the receiver input is 10^{-11} W/Hz. The probability of error for an optimum filter is _____. | 1 | 3 | 4 |
| (A) $0.5 \operatorname{erfc}(\sqrt{5.5})$ | (B) $\operatorname{erfc}((5.5))$ | | |
| (C) $0.5 \operatorname{erfc}(\sqrt{5})$ | (D) $\operatorname{erfc}(\sqrt{5})$ | | |
| 17. The capacity of a noise-free channel having M symbols is _____. | 1 | 2 | 5 |
| (A) M | (B) 2M | | |
| (C) $\log M$ | (D) infinite | | |
| 18. The entropy of a source generating 3 messages with probabilities 0.5, 0.25, and 0.25 is _____ per message. | 1 | 3 | 5 |
| (A) 1 | (B) 0.33 | | |
| (C) 1.5 | (D) 3 | | |
| 19. An extremely noisy channel has a bandwidth of 64 KHz. What is the capacity of this channel? | 1 | 2 | 5 |
| (A) 32 Kbps | (B) 8 Kbps | | |
| (C) 4 Kbps | (D) Nearly 0 bps | | |
| 20. The efficiency of Huffman code is linearly proportional to _____. | 1 | 4 | 5 |
| (A) number of symbols | (B) maximum length of the code | | |
| (C) average length of the code | (D) average entropy | | |

PART - B (5 × 8 = 40 Marks)

Answer all Questions

Marks BL CO

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|--|---|---|---|
| 21. (a) With neat circuit diagram, explain the operation of the collector modulator. | 8 | 2 | 1 |
| (OR) | | | |
| (b) With relevant circuit diagram and equations, show how a Foster-Seeley discriminator extract the modulating signal. | | | |
| 22. (a) Draw the block diagram of AM superheterodyne receiver and emphasize the contribution of each block in achieving the best performance. | 8 | 2 | 2 |
| (OR) | | | |
| (b) Derive the general expression of Figure of merit for a FM system and show that it is 1.5 times the square of modulation index for a single tone modulating signal. | | | |

23. (a) Draw the block diagram of a PCM system with relevant waveform illustrations for each block. Also explain how adaptive step size is achieved using Companding. 8 2 3

(OR)

- (b) What is inter symbol interference? How does it differ from crosstalk? Give a quality analysis based on possible inference gained from eye patterns.
24. (a) Briefly explain the the generation and detection of DPSK signals with relevant sketches. 8 2 4

(OR)

- (b) Briefly explain the the generation and detection of QPSK signals with relevant sketches.
25. (a) i) State the Shannon-Hartley capacity theorem of a Gaussian channel and obtain the Shannon capacity limit. (4 Marks) 8 3 5
ii) For a voice communication channel of transmission bandwidth 3.4 KHz with SNR of 18 dB, deduce the channel capacity and its Shannon limit. (4 Marks)

(OR)

- (b) A information source produces 8 symbols $\{x_1, x_2, x_3, x_4, x_5, x_6, x_7, \text{ and } x_8\}$ with probabilities $\{1/4, 1/8, 1/16, 1/16, 1/16, 1/4, 1/16, \text{ and } 1/8\}$ respectively.
i) Apply Shannon-Fano coding encoding. (4 Marks)
ii) Find its efficiency for $M=2$. (4 Marks)

PART - C ($1 \times 15 = 15$ Marks)

Marks BL CO

Answer any 1 Questions

26. Derive the following expressions for a binary PAM system: 15 3 3
i) Maximum output SNR of optimum filter. (6 Marks)
ii) Impulse response of matched filter. (3 Marks)
iii) Error probability of matched filter. (6 Marks)
27. Explain the following M-ary digital modulation schemes with relevant equations and signal space diagrams: (5 Marks each) 15 3 4
i) 8-ary PSK
ii) 16-ary QAM
iii) 3-ary FSK

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